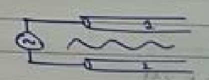
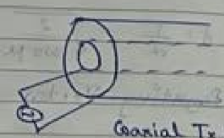
Transmission Lines

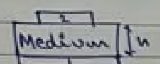
- 2 conducting system with finite gap.
- Source is applied b/w 2 conduc.



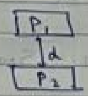
2 wire Tx line.



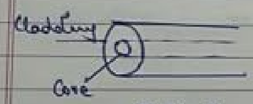
Coaxial Tx line.



Microchip Tx line

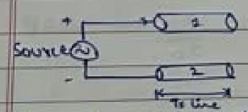


Planar or Parallel plate Tx line.



Optical fibre

Equivalent Circuit

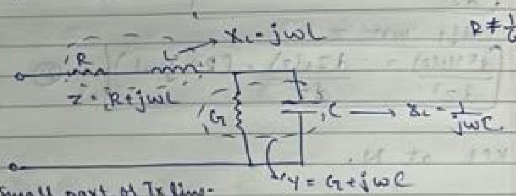


(i) Due to current through Tx line,
- mag field is produced
- " flux is produced. → flux linkage
- represented by Inductance. L

(ii) Conductors are not perfect.
- Ohmic losses.
- represented by resistance R

(iii) Conductors separated by medium.
- dielectric losses.
- represented by conductance (G).

(iv) Conductors separated by distance.
- dielectric is present.
- capacitance C



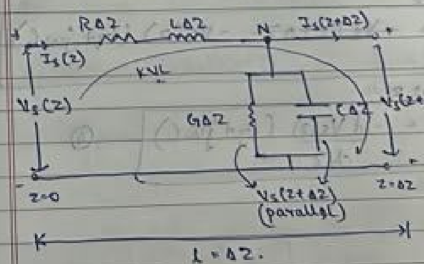
- small part of Tx line -

Primary Constants: R, L, G, C.

Transmission Line Equation

Considering small length,

Δz = small length
R = res per unit length
$R\Delta z = \Omega$



$V_s(z) = I_s(z)R\Delta z + I_s(z)j\omega L\Delta z + V_s(z+\Delta z)$.

$V_s(z+\Delta z) - V_s(z) = -I_s(z)\Delta z(R+j\omega L)$.

$\dfrac{V_s(z+\Delta z) - V_s(z)}{\Delta z} = -I_s(z)(R+j\omega L)$.

$\lim\limits_{\Delta z\to 0} \dfrac{V_s(z+\Delta z) - V_s(z)}{\Delta z} = \lim\limits_{\Delta z\to 0} -I_s(z)(R+j\omega L)$.

$$\boxed{\frac{dV_s(z)}{dz} = -I_s(z)(R+j\omega L)} \quad (1)$$

diff wrt z.

$$\boxed{\frac{d^2V_s(z)}{dz^2} = -\frac{dI_s(z)}{dz}\cdot(R+j\omega L)} \quad (2)$$

KCL at N,

$I_s(z) = I_s(z+\Delta z) + V_s(z+\Delta z)G\Delta z + \dfrac{V_s(z+\Delta z)}{1/\Delta z j\omega C}$.

$I_s(z) = I_s(z+\Delta z) + V_s(z+\Delta z)G\Delta z + V_s(z+\Delta z)C\Delta z j\omega$.

$-V_s(z+\Delta z)G\Delta z - V_s(z+\Delta z)C\Delta z j\omega = I_s(z+\Delta z) - I_s(z)$.

$I_s(z+\Delta z) - I_s(z) = -V_s(z+\Delta z)\Delta z(G+Cj\omega)$.

$\dfrac{I_s(z+\Delta z) - I_s(z)}{\Delta z} = -V_s(z+\Delta z)(G+Cj\omega)$.

$\lim\limits_{\Delta z\to 0} \dfrac{I_s(z+\Delta z) - I_s(z)}{\Delta z} = \lim\limits_{\Delta z\to 0} -V_s(z+\Delta z)(G+Cj\omega)$

$$\boxed{\frac{dI_s(z)}{dz} = \cancel{-V_s(z+\Delta z)} - V_s(z)(G+j\omega C)} \quad (3)$$

diff wrt z.

$$\boxed{\frac{d^2I_s(z)}{dz^2} = -\frac{dV_s(z)}{dz}(G+j\omega C)} \quad (4)$$

(3) in (2),

$\dfrac{d^2V_s(z)}{dz^2} = +V_s(z)(G+j\omega C)(R+j\omega L)$.

$\dfrac{d^2V_s(z)}{dz^2} - \underbrace{(G+j\omega C)(R+j\omega L)}_{\gamma^2}V_s(z) = 0$

$$\boxed{\frac{d^2V_s(z)}{dz^2} - \gamma^2 V_s(z) = 0}$$ Tx line eqn. (Voltage wave).

~~$\gamma = R+j\omega L$~~ $\gamma = \sqrt{(R+j\omega L)(G+j\omega C)}$

↳ Propagation constant.

~~(4) in (1)~~, (1) in (4),

$\dfrac{d^2I_s(z)}{dz^2} = +I_s(z)(R+j\omega L)(G+j\omega C)$.

$\dfrac{d^2I_s(z)}{dz^2} - (R+j\omega L)(G+j\omega C)I_s(z) = 0$

$$\boxed{\frac{d^2I_s(z)}{dz^2} - \gamma^2 I_s(z) = 0}$$ Tx line eqn (Current wave).

Solution of Equation

$\dfrac{d^2V_s(z)}{dz^2} - \gamma^2 V_s(z) = 0$ $\quad \dfrac{d}{dz} \to D \quad \dfrac{d^2}{dz^2} \to D^2$

$(D^2 - \gamma^2)V_s(z) = 0$

$D^2 - \gamma^2 = 0$

$D^2 = \gamma^2$

$D = \pm\gamma$

m_1, m_2 roots.

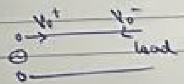
* $y(t) = \underbrace{C_1}e^{m_1 t} + \underbrace{C_2}_{\text{constant}} e^{m_2 t}$

∵ $\boxed{\text{imp' mismatch}}$

$$\boxed{V_s(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}}$$

Forward travelling vtg (+z axis).

Reverse travelling vtg. (-z axis)



$$\frac{d^2 I_s(z)}{dz^2} - \gamma^2 I_s(z) = 0.$$

$$(D^2 - \gamma^2)\, I_s(z) = 0.$$

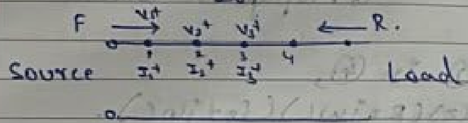
$$D^2 = \gamma^2$$

$$D = \pm\gamma$$

$$\boxed{I_s(z) = I_0^+ e^{-\gamma z} + I_0^- e^{+\gamma z}}$$

Characteristic Impedance (Z_0).

ratio of $\frac{V_0^+}{I_0^+} = Z_0^+$ (ohm).



$$Z_0 = \frac{V_1^+}{I_1^+} = \frac{V_2^+}{I_2^+} = \frac{V_3^+}{I_3^+}$$

V = exponential decay

I =

$$\boxed{Z_0 = \frac{V_0^+}{I_0^+} = -\frac{V_0^-}{I_0^-}} \quad \text{—①}$$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \quad \text{—②}$$

Lossless Tx line

Condition: $R = 0$, $G = 0$.

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$\alpha + j\beta = \sqrt{(j\omega L)(j\omega C)} = j\omega\sqrt{LC}$$

Comparing,

$\alpha = 0$; $\beta = \omega\sqrt{LC}$

attenuation constant; phase constant.

Phase velocity, $\boxed{V_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}}}$

Characteristic Imp. $Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \Rightarrow \boxed{Z_0 = \sqrt{\frac{L}{C}}}$

$$\beta Z_0 = \omega\sqrt{LC} \cdot \sqrt{\frac{L}{C}}$$

$$\beta Z_0 = \omega L \qquad \boxed{L = \frac{\beta Z_0}{\omega}}$$

$$\beta \times V_p = \omega\sqrt{LC} \times \frac{1}{\sqrt{LC}} \qquad \boxed{\beta V_p = \omega}$$

$$\frac{\beta}{Z_0} = \omega\sqrt{LC} \times \sqrt{\frac{C}{L}}$$

$$\boxed{C = \frac{\beta}{Z_0\,\omega}}$$

Distortionless Tx line

signals with diff freqs reach load at same time.

Condition: $\boxed{\frac{R}{L} = \frac{G}{C}}$

$\gamma = \alpha + j\beta = \sqrt{(R+j\omega L)(G+j\omega C)}$

$\alpha + j\beta = \sqrt{RG\left(1+\frac{j\omega L}{R}\right)\left(1+\frac{j\omega C}{G}\right)}$

use $\frac{C}{G} = \frac{L}{R}$.

$\alpha + j\beta = \sqrt{RG\left(1+j\omega\frac{L}{R}\right)^2} = \sqrt{RG} \times \left(1+\frac{j\omega L}{R}\right)$

$= \sqrt{RG} + j\omega \frac{L}{R}\sqrt{RG} = \sqrt{RG} + j\omega\sqrt{\frac{L^2}{R^2} \times R \times G}$ ... $\sqrt{\ldots R \times \frac{C^2}{G^2}}$

$= \sqrt{RG} + j\omega\sqrt{\frac{L}{\cancel{C}} \times C^2}$

$\boxed{\alpha + j\beta = \sqrt{RG} + j\omega\sqrt{LC}}$

$\boxed{\alpha = \sqrt{RG} \quad \beta = \omega\sqrt{LC}}$

$Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}} = \sqrt{\frac{R(1+j\omega L/R)}{G(1+j\omega C/G)}}$

use $\frac{C}{G} = \frac{L}{R}$.

$\boxed{Z_0 = \sqrt{\frac{R}{G}} \qquad Z_0 = \sqrt{\frac{L}{C}}}$

$\frac{R}{G} = \frac{L}{C}$

$V_p = \frac{\omega}{\beta} = \frac{\omega}{\omega\sqrt{LC}} = \boxed{\frac{1}{\sqrt{LC}} = V_p}$

$\alpha Z_0 = \sqrt{RG} \times \sqrt{\frac{L}{C}} = \sqrt{RG} \times \sqrt{\frac{R}{G}}$

$\boxed{\alpha Z_0 = R}$

$\frac{\alpha}{Z_0} = \sqrt{RG} \times \sqrt{\frac{C}{L}} = \sqrt{RG} \times \sqrt{\frac{G}{R}} = G$

$\boxed{\frac{\alpha}{Z_0} = G}$

$\beta Z_0 = \omega\sqrt{L\cancel{C}} \times \sqrt{\frac{L}{\cancel{C}}} = \boxed{\omega L = \beta Z_0}$

$\boxed{L = \frac{\beta Z_0}{\omega}}$

$\frac{\beta}{Z_0} = \omega\sqrt{\cancel{L}C} \times \sqrt{\frac{C}{\cancel{L}}} = \omega C$

$\boxed{C = \frac{\beta}{Z_0\omega}}$

lossless	distortionless
$\alpha = 0$	$\alpha = \sqrt{RG}$

Low-loss Tx line

Condition: $R \ll j\omega L$
$G \ll j\omega C$

$\Rightarrow RG \ll j^2\omega^2 LC$

$\gamma = \alpha + j\beta = \sqrt{(R+j\omega L)(G+j\omega C)}$

$$\alpha + j\beta = \sqrt{j\omega L\,(1 + R/j\omega L)\; j\omega C\,(1 + G/j\omega C)}$$

$$= j\omega\sqrt{LC}\left(1 + \frac{R}{j\omega L}\right)^{1/2}\left(1 + \frac{G}{j\omega C}\right)^{1/2}$$

$$= j\omega\sqrt{LC}\left(1 - \frac{R}{\omega L}\right)^{1/2}\left(1 - \frac{G}{\omega C}\right)^{1/2}$$

use $\boxed{(1+x)^n = 1 + nx}$ $x \ll 1$

$$\alpha + j\beta = j\omega\sqrt{LC}\left(1 - \frac{jR}{2\omega L}\right)\left(1 - \frac{jG}{2\omega C}\right)$$

$$\alpha + j\beta = j\omega\sqrt{LC}\left(1 - \frac{jR}{2\omega L} - \frac{jG}{2\omega C} - \underbrace{\frac{RG}{4\omega^2 LC}}_{\approx 0}\right)$$

$$\alpha + j\beta = j\omega\sqrt{LC}\left(1 - \frac{jR}{2\omega L} - \frac{jG}{2\omega C}\right)$$

$$= j\omega\sqrt{LC} + \frac{R\omega\sqrt{LC}}{2\omega L} + \frac{G\omega\sqrt{LC}}{2\omega C}$$

$$= j\omega\sqrt{LC} + \frac{R}{2}\sqrt{\frac{C}{L}} + \frac{G}{2}\sqrt{\frac{L}{C}}$$

Comparing

$$\boxed{\alpha = \frac{R}{2}\sqrt{\frac{C}{L}} + \frac{G}{2}\sqrt{\frac{L}{C}}} \qquad \boxed{\beta = \omega\sqrt{LC}}$$

$$v_p = \frac{\omega}{\beta} = \frac{\omega}{\omega\sqrt{LC}} \qquad \boxed{v_p = \frac{1}{\sqrt{LC}}}$$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{j\omega L\,(1 + R/j\omega L)}{j\omega C\,(1 + G/j\omega C)}}$$

$$Z_0 = \sqrt{\frac{L}{C}}\left(1 + \frac{R}{j\omega L}\right)^{1/2}\left(1 + \frac{G}{j\omega C}\right)^{-1/2}$$

$$= \sqrt{\frac{L}{C}}\left(1 - \frac{Rj}{\omega L}\right)^{1/2}\left(1 - \frac{Gj}{j\omega C}\right)^{-1/2}$$

$$= \sqrt{\frac{L}{C}}\left(1 - \frac{Rj}{2\omega L}\right)\left(1 + \frac{Gj}{2\omega C}\right)$$

$$= \sqrt{\frac{L}{C}}\left(1 - \frac{Rj}{2\omega L} + \frac{Gj}{2\omega C} + \underbrace{\frac{RG}{4\omega^2 LC}}_{\approx 0}\right)$$

$$Z_0 = \underbrace{\sqrt{\frac{L}{C}}}_{\text{real}} \underbrace{- \frac{R}{2\omega L}\sqrt{\frac{L}{C}}\,j + \frac{G}{2\omega C}\sqrt{\frac{L}{C}}\,j}_{\text{img.}}$$

if Z_0 is real, $\text{img}(Z_0) = 0$.

$$\frac{G}{2\omega C}\sqrt{\frac{L}{C}} = \frac{R}{2\omega L}\sqrt{\frac{L}{C}}$$

$$\boxed{\frac{G}{C} = \frac{R}{L}} \Rightarrow \text{Distortionless.}$$

Z_0 is given & real $\Rightarrow$ Tx is ~~d~~

Problems

L, C = ?

① Air medium, $Z_0 = 50\,\Omega$, $\beta = 3$ rad/m at 150 MHz

Ans $\hookrightarrow$ loss less

$R = G = 0.$

$Z_0 = 50$ $\qquad \beta = 3\ \text{rad m}^{-1}.$

$f = 150\ \text{MHz} \Rightarrow \omega = 2\pi f$

$$L = \frac{\beta Z_0}{\omega} = \frac{3 \times 50}{2\pi \times 150 \times 10^6} = \frac{1}{2\pi} \times 10^{-6}\ \text{H/m.}$$

$$C = \frac{\omega}{\beta}\ \frac{\beta}{Z_0\omega} = \frac{3}{2\pi \times 150 \times 10^6 \times 50} = \frac{10^{-6}}{2500 \times 2\pi}\ \text{F/m.}$$

② $f = 600\,MHz$ $Z_0 = 50\Omega$ $\alpha = 0.04$

$\beta = 2\,rad\,m^{-1}$ find R, G, L, C.

Ans $Z_0 = 50\Omega$ (Real)

$\Longrightarrow$ Distortionless. $\frac{R}{L} = \frac{G}{C}$

$\alpha = \sqrt{RG}$ $\quad Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{R}{G}}$

$R = \alpha Z_0 = 0.04 \times 50$

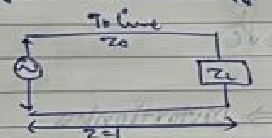
$\boxed{R = 2\,\Omega\,m^{-1}}$

$G = \frac{\alpha}{Z_0} = \frac{0.04}{50}$

$L = \frac{\beta Z_0}{\omega} = \frac{2 \times 50}{2\pi \times 600 \times 10^6}$ $H\,m^{-1}$

$C = \frac{\beta}{Z_0 \omega} = \frac{2}{2\pi \times 600 \times 10^6 \times 50}$ F/m

Reflection Co-eff



$V_s(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$

$I_s(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}$

at $z=0$,

$V_s(0) = V_0^+ + V_0^-$

$I_s(0) = I_0^+ + I_0^-$

$Z_L = \frac{V(z=0)}{I(z=0)} = \frac{V_0^+ + V_0^-}{I_0^+ + I_0^-}$

use $Z_0 = \frac{V_0^+}{I_0^+} = \frac{-V_0^-}{I_0^-}$

$I_0^+ = \frac{V_0^+}{Z_0}$, $I_0^- = \frac{-V_0^-}{Z_0}$

$Z_L = \frac{V_0^+ + V_0^-}{\frac{V_0^+}{Z_0} - \frac{V_0^-}{Z_0}}$ $\quad Z_L = \frac{(V_0^+ + V_0^-)}{(V_0^+ - V_0^-)} \cdot Z_0$

$= \frac{V_0^+(1 + V_0^-/V_0^+)}{V_0^+(1 - V_0^-/V_0^+)}$

where $\boxed{\frac{V_0^-}{V_0^+} = \Gamma}$ ← ref-coeff

$\therefore \boxed{Z_L = \left(\frac{1+\Gamma}{1-\Gamma}\right) Z_0}$

$\cancel{Z_L\Gamma - Z_L}$ $\quad Z_L - Z_L\Gamma = Z_0 + Z_0\Gamma$

$\boxed{\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}}$

if $Z_L = Z_0$

$\Rightarrow \Gamma = 0$.

else $0 \le \Gamma \le 1$.

Transmission Co-eff (T)

$\cancel{\boxed{T + \Gamma = 0}}$ $\quad \boxed{T + \Gamma = 1}$ **

$\boxed{T = \frac{2Z_L}{Z_L + Z_0}}$

Voltage Standing Wave Ratio (VSWR)

VSWR → S.

$= \frac{V_{max}}{V_{min}} = \frac{V^+ + V^-}{V^+ - V^-} = S$

$S = \frac{V^+(1 + V^-/V^+)}{V^+(1 - V^-/V^+)}$

WKT. $\frac{V_0^-}{V_0^+} = \Gamma$

$$\boxed{S = \frac{1+\Gamma}{1-\Gamma}}$$

if $\Gamma = 0$, $S = 1$.
if $\Gamma = 1$, $S = \infty$

$$\boxed{1 \leq S < \infty}$$

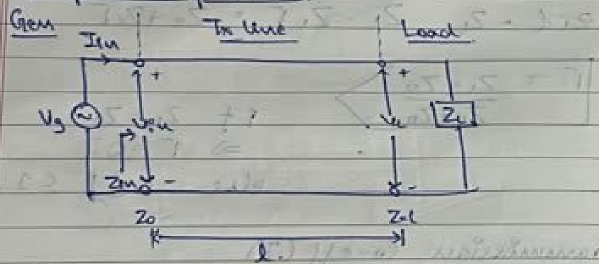
$$Z_{max} = \frac{V_{max}}{I_{min}} = \frac{V_{max}}{V_{min}} \times \frac{V_{min}}{I_{min}}$$

$$\boxed{Z_{max} = S Z_0}$$

$$Z_{min} = \frac{V_{min}}{I_{max}} = \frac{V_{min}}{V_{max}} \times \frac{V_{max}}{I_{max}}$$

$$\boxed{Z_{min} = \frac{1}{S} Z_0}$$

Input Impedence



$$V_s(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$$
$$I_s(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}.$$

$$\frac{V_0^+}{I_0^+} = Z_0 \qquad -\frac{V_0^-}{I_0^-} = Z_0.$$

At $z = l$

$$V_s(z=l) = V_0^+ e^{-\gamma l} + V_0^- e^{\gamma l} = V_L \text{ ——— } ①$$
$$I_s(z=l) = I_0^+ e^{-\gamma l} + I_0^- e^{\gamma l} = I_L$$
$$= \frac{V_0^+}{Z_0} e^{-\gamma l} - \frac{V_0^-}{Z_0} e^{\gamma l} = I_L$$

$$I_L Z_0 = V_0^+ e^{-\gamma l} - V_0^- e^{\gamma l} \text{ ——— } ②$$

① + ②

$$2V_0^+ e^{-\gamma l} = V_L + I_L Z_0$$
$$2V_0^+ e^{-\gamma l} = I_L Z_L + I_L Z_0.$$
$$\boxed{V_0^+ = \frac{(I_L Z_L + I_L Z_0) e^{\gamma l}}{2}}$$

① − ②

$$2V_0^- e^{\gamma l} = V_L - I_L Z_0$$
$$2V_0^- e^{\gamma l} = I_L Z_L - I_L Z_0$$
$$\boxed{V_0^- = \frac{(I_L Z_L - I_L Z_0) e^{-\gamma l}}{2}}$$

At $z = 0$, $V_s(0) = V_0^+ + V_0^-$
$I_s(0) = I_0^+ + I_0^- = \frac{V_0^+ - V_0^-}{Z_0}$

$$Z_{in} = \frac{V_{in}}{I_{in}} = \frac{V_s(0)}{I_s(0)} = \frac{(V_0^+ + V_0^-) \cdot Z_0}{(V_0^+ - V_0^-)}$$
$$= Z_0 \left(\frac{V_0^+ + V_0^-}{V_0^+ - V_0^-} \right) \text{ ——— } ③$$

$$Z_{in} = Z_0 \left\{ \frac{\left(\frac{I_L Z_L + I_L Z_0}{2}\right) e^{\gamma l} + \left(\frac{I_L Z_L - I_L Z_0}{2}\right) e^{-\gamma l}}{\left(\frac{I_L Z_L + I_L Z_0}{2}\right) e^{\gamma l} - \left(\frac{I_L Z_L - I_L Z_0}{2}\right) e^{-\gamma}} \right\}$$

↳ I_L common from all terms ⇒ cancel.

$$Z_{in} = \left\{ \frac{\frac{Z_L e^{\gamma l}}{2} + \frac{Z_0 e^{\gamma l}}{2} + \frac{Z_L e^{-\gamma l}}{2} - \frac{Z_0 e^{-\gamma l}}{2}}{\frac{Z_L e^{\gamma l}}{2} + \frac{Z_0 e^{\gamma l}}{2} - \frac{Z_L e^{-\gamma l}}{2} + \frac{Z_0 e^{-\gamma l}}{2}} \right\} Z_0.$$

$$Z_{in} = Z_0 \left\{ \frac{Z_L \left(\frac{e^{\gamma l} + e^{-\gamma l}}{2}\right) + Z_0 \left(\frac{e^{\gamma l} - e^{-\gamma l}}{2}\right)}{Z_L \left(\frac{e^{\gamma l} - e^{-\gamma l}}{2}\right) + Z_0 \left(\frac{e^{\gamma l} + e^{-\gamma l}}{2}\right)} \right\}.$$

$$Z_{in} = Z_0 \left\{ \frac{Z_L \cosh \gamma l + Z_0 \sinh \gamma l}{Z_0 \cosh \gamma l + Z_L \sinh \gamma l} \right\}.$$

Take $\cosh \gamma l$ common.

$$Z_{in} = Z_0 \left[\frac{Z_L + Z_0 \tanh \gamma l}{Z_0 + Z_L \tanh \gamma l}\right]$$

For lossless.

$\gamma = \alpha + j\beta$ $\alpha = 0$

$\gamma = j\beta$.

$$\tanh \gamma l = \frac{\left(\frac{e^{\gamma l} - e^{-\gamma l}}{2}\right)}{\left(\frac{e^{\gamma l} + e^{-\gamma l}}{2}\right)}$$

$\gamma = j\beta$

$$= j\frac{\left(\frac{e^{j\beta l} - e^{-j\beta l}}{2j}\right)}{\left(\frac{e^{j\beta l} + e^{-j\beta l}}{2}\right)}$$

$$= j\frac{\sin \beta l}{\cos \beta l} = j \tan \beta l.$$

$$Z_{in} = Z_0 \left[\frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l}\right]$$

βl = Electrical length. (rad).

(i) if $l = \frac{\lambda}{8}$, $Z_{in} = ?$

$\beta = \frac{2\pi}{\lambda}$

$$\beta l = \frac{2\pi}{\lambda} \times \frac{\lambda}{8} = \frac{\pi}{4}.$$

$\tan \beta l = 1$.

$$Z_{in} = Z_0 \left[\frac{Z_L + jZ_0}{Z_0 + jZ_L}\right]$$

$x = a + jb$

$|x| = \sqrt{a^2 + b^2}$

$$|Z_{in}| = Z_0 \left\{\frac{\sqrt{Z_L^2 + Z_0^2}}{\sqrt{Z_0^2 + Z_L^2}}\right\}$$

$|Z_{in}| = Z_0$ $\quad l = \frac{\lambda}{8}$

(ii) $l = \lambda/4$

$\beta l = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{2}$ $\quad \tan \beta l = \infty$

$$Z_{in} = Z_0 \left[\frac{\tan \beta l \left(\cancel{\frac{Z_L}{\tan \beta l}}^{0} + jZ_0\right)}{\tan \beta l \left(\cancel{\frac{Z_0}{\tan \beta l}}^{0} + jZ_L\right)}\right].$$

$$Z_{in} = Z_0 \left[\frac{jZ_0}{jZ_L}\right] = \frac{Z_0^2}{Z_L}$$

$$\boxed{Z_{in} = \frac{Z_0^2}{Z_L}}$$

$Z_{in} \propto \frac{1}{Z_L}$

if $Z_L = \infty \Rightarrow Z_{in} = 0$ (open circuit)

if $Z_L = 0 \Rightarrow Z_{in} = \infty$ (short circuit).

(iii) $l = \lambda/2 \Rightarrow \beta l = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi \Rightarrow \tan \pi = 0$.

$$Z_{in} = Z_0 \left[\frac{Z_L}{Z_0}\right]$$

$$\boxed{Z_{in} = Z_L}$$

iv for matched load

$Z_L = Z_0$

$$Z_{in} = Z_0 \left[\frac{Z_0 + jZ_0 \tan\beta l}{Z_0 + jZ_0 \tan\beta l} \right]$$

$\boxed{Z_{in} = Z_0} \implies \Gamma = 0$

SWR = absent.

v load imp is short circuit → Z_{sc}.

$Z_L = 0\,\Omega$.

$$Z_{in} = Z_0 \left[\frac{0 + jZ_0 \tan\beta l}{Z_0 + 0} \right]$$

$\boxed{Z_{in} = jZ_0 \tan\beta l}$

$Z_{sc} = jZ_0 \tan\beta l$.

→ $0 < l < \lambda/4$. $\quad \beta l = \frac{2\pi}{\lambda} \times \frac{\lambda}{4}$

$0 < \beta l < \frac{\pi}{2}$

$\tan\beta l = +ve$. $\quad \tan\beta l = +k$.

$Z_{sc} = jZ_0 k \longrightarrow$ purely inductive. ($\because j$ is present)

→ $\lambda/4 < l < \lambda/2$. $\quad \beta l = \frac{2\pi}{\lambda} \times \frac{\lambda}{2}$

$\frac{\pi}{2} < \beta l < \pi$

$\tan\beta l = -ve$ $\quad \tan\beta l = -k$.

$Z_{sc} = -jZ_0 k \longrightarrow$ purely capacitive ($\because -j$ is present)

→ $\frac{\lambda}{2} < l < \frac{3\lambda}{4}$

$\pi < \beta l < \frac{3\pi}{2}$

$\tan\beta l = +ve$

$Z_{sc} = jZ_0 k \rightarrow$ Inductive

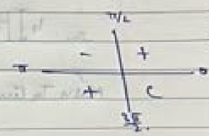
→ III^{ly} $\frac{3\lambda}{4} < l < \lambda$ capacitive.

vi load imp is open circuit.

$Z_L = \infty\,\Omega$.

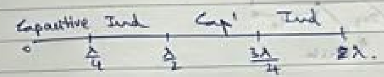
$$Z_{in} = Z_0 \left(\frac{Z_L \left(1 + \frac{jZ_0 \tan\beta l}{Z_L}\right)}{Z_L \left(\frac{Z_0}{Z_L} + j\tan\beta l\right)} \right)$$

$$= \frac{Z_0}{j\tan\beta l}.$$



$\boxed{Z_{in} = -jZ_0 \cot\beta l}$ $\quad Z_{oc} = -jZ_0 \cot\beta l$

Same cases as before



※ attenuation α = np/m

to dB, $\boxed{\alpha \times 8.686}$

Stub Matching

It is SC or OC Tx line

Why stub?

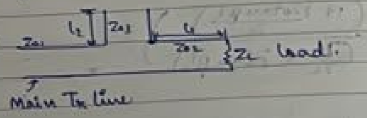
$Z_L \neq Z_0 \Rightarrow$ reflection occurs.

If $\Gamma > 0$, max power will not be transferred to load.

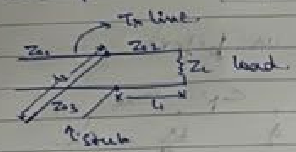
To match Z_L & Z_0, we use stub in Tx line

- Series connection (z)
- Parallel connection (y).

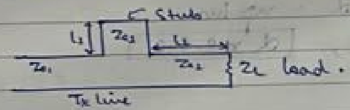
→ Open Ckt series stub.



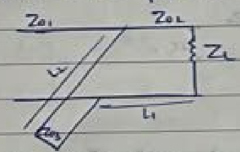
→ Open ckt parallel stub



→ Short ckt series stub.

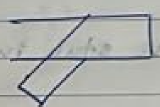


→ Short ckt parallel stub.



Short Circuit stub is better than open.

Types



Single

Double.

Normalised load imp

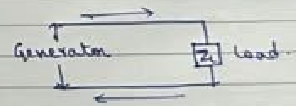
$\overline{Z_L} = \frac{Z_L}{Z_0}$

Normalised input imp

$\overline{Z_{in}} = \frac{Z_{in}}{Z_0}$

SMITH CHART

- Graphical method.
- To find Γ, Z_{in}, VSWR, stub matching.
- Types: Constant resistance circle
  Constant reactance circle
- Works on normalised imp.
- One complete circle = $\lambda/2$ distance.
-



load to generator : move clockwise.
generator to load : move anti-clock